

ImageURI Field

This field, when present, SHALL contain a URI where the OTA Requestor SHOULD download a Software Image. The syntax of the ImageURI field SHALL follow the URI syntax as specified in [RFC 3986](#).

Beware, this field is conditionally present based on the conformance listed in [Section 11.20.6.5.2, “QueryImageResponse Command”](#).

If the ImageURI specifies a [BDX Protocol bdx:](#) scheme, then the following rules describe the location to be used for download:

1. The URI's [scheme](#) field SHALL be exactly [bdx](#) in lowercase characters.
2. The URI's [authority](#) field SHALL contain only the [host](#) portion and SHALL use string representation of the [Operational Node ID](#) of the Node where to proceed with the download, on the same Fabric on which the OTA Requestor received the QueryImageResponse.
3. The encoding of the Node ID in the [host](#) field SHALL use an uppercase hexadecimal format, using exactly 16 characters to encode the network byte order value of the NodeID, in a similar fashion as the Node Identifier portion of the [Operational Instance Name](#).
 - a. The Operational Node ID in the [host](#) field SHALL match the NodeID of the OTA Provider responding with the QueryImageResponse. The usage of a different Node ID than that of the provider is reserved for future use. This constraint reduces the number of independent [CASE](#) secure channel sessions that have to be maintained to proceed with OTA software updates, thus reducing energy and resource utilization for the software update process.
4. The [user](#) section of the [authority](#) field SHALL be absent, as there are no "users" to be considered.
5. The [port](#) section of the [authority](#) field SHALL be absent, as the port for transport SHALL be determined through [Operational Discovery](#) of the target Node.
6. The URI SHALL NOT contain a [query](#) field.
7. The URI SHALL NOT contain a [fragment](#) field.
8. The [path](#) field SHALL employ absolute path representation and SHALL contain the [file designator](#) of the software image to download at the BDX server. When used with the BDX server, the leading [/](#) separating the URI authority from the path SHALL be omitted. When contacting the BDX server, further processing of the file designator SHALL NOT be done, including handling of URL-encoded escape sequences. Rather, the exact octets of the path, as received SHALL be the values used by both client and server in handling the file designator.
 - a. The [path](#) SHALL only contain valid URI characters.

These rules above for BDX URIs simplify parsing for OTA Requestors receiving Image URIs. The following example procedure shows how the format constraints simplify the extraction of the necessary data to reach the BDX server for download.

1. Verify that the URI is 24 characters or longer, which is the minimum length of a valid BDX URI with all elements present, for example [bdx://00112233AABBCCDD/0](#).
2. Verify the presence of prefix [bdx://](#) indicating a BDX URI.
3. Extract the next 16 characters and convert from uppercase hexadecimal to a 64-bit scalar value,